EMMANUEL OLUKAYEGUN
OLATEJU
CEID: eolateju
STUDENT ID : 1984858

Deep RL for ROBOTICS
ASSIGNMENT TWO

3.] THEORETICAL FOUNDATIONS OF PPO

Q1.] Generalized Advantage Estimation (GAE) and Bias-Variance Trade-off

→ 1-Step TD Residual:

$$\delta_t^V = r_t + \gamma V(s_{t+1}) - V(s_t)$$

→ K-Step Advantage Estimation

$$\hat{A}_t^{(k)} = \sum_{l=0}^{k-1} \gamma^l \delta_{t+l}^V = -V(s_t) + \sum_{l=0}^{k-1} \gamma^l r_{t+l} + \gamma^k V(s_{t+k})$$

→ Generalized Advantage Estimation

$$\hat{A}_t^{GAE(\gamma,\lambda)} = (1-\lambda) \sum_{k=1}^{\infty} \lambda^{k-1} \hat{A}_t^{(k)} \; ; \quad \lambda \in [0,1]$$

(a.) Prove by algebraic substitution and grouping of powers γ and λ that:

$$\hat{A}_t^{GAE(\gamma,\lambda)} = \sum_{l=0}^{\infty} (\gamma\lambda)^l \delta_{t+l}^V$$

$$\hat{A}_t^{GAE(\gamma,\lambda)} = (1-\lambda)\left[\lambda^0 \hat{A}_t^{(0)} + \lambda^1 \hat{A}_t^{(1)} + \lambda^2 \hat{A}_t^{(2)} + \cdots\right]$$

$$= (1-\lambda)\left[\lambda^0(\gamma^0 \delta_{t+0}^V) + \lambda^1(\gamma^0 \delta_{t+0}^V + \gamma^1 \delta_{t+1}^V) + \lambda^2(\gamma^0 \delta_{t+0}^V + \gamma^1 \delta_{t+1}^V + \gamma^2 \delta_{t+2}^V) + \cdots\right]$$

$$= (1-\lambda)\left[\gamma^0 \delta_{t+0}^V(\lambda^0 + \lambda^1 + \lambda^2 + \cdots) + \lambda^1 \gamma^1 \delta_{t+1}^V(\lambda^0 + \lambda^1 + \cdots) + \lambda^2 \gamma^2 \delta_{t+2}^V(\lambda^0 + \lambda^1 + \lambda^2 + \cdots) + \cdots\right]$$

$$\hat{A}_t^{GAE(\gamma,\lambda)} = (1-\lambda)\left[\lambda^0\gamma^0\delta_{t+0}^V \sum_{m=0}^{\infty}\lambda^m + \lambda^1\gamma^1\delta_{t+1}^V \sum_{m=0}^{\infty}\lambda^m + \lambda^2\gamma^2\delta_{t+2}^V \sum_{m=0}^{\infty}\lambda^m + \cdots\right]$$

$$\sum_{m=0}^{\infty}\lambda^m = \frac{1}{1-\lambda} \quad : |\lambda|<1$$

$$\therefore \hat{A}_t^{GAE(\gamma,\lambda)} = \left(\lambda^0\gamma^0\cdot\delta_{t+0}^V\cdot\frac{1}{1-\lambda} + \lambda^1\gamma^1\cdot\delta_{t+1}^V\cdot\frac{1}{1-\lambda} + \lambda^2\gamma^2\cdot\delta_{t+2}^V\cdot\frac{1}{1-\lambda} + \cdots\right)(1-\lambda)$$

$$\boxed{\hat{A}_t^{GAE(\gamma,\lambda)} = \sum_{l=0}^{\infty}(\lambda\gamma)^l\delta_{t+l}^V}$$

(b) Show $\hat{A}_t^{GAE(\gamma,\lambda)} = \delta_t^V$ when $\lambda = 0$

$$\hat{A}_t^{GAE(\gamma,\lambda=0)} = \hat{A}_t^{GAE(\gamma,0)}$$

$$= \sum_{l=0}^{\infty}\gamma^l\cdot(0)^l\cdot\delta_{t+l}^V$$

$$= \gamma^0\cdot(0)^0\cdot\delta_{t+0}^V + \underbrace{\sum_{l=1}^{\infty}\gamma^l\cdot(0)^l\cdot\delta_{t+l}^V}_{\to 0}$$

$$= (1)\cdot(1)\cdot\delta_{t+0}^V$$

$$\therefore \boxed{\hat{A}_t^{GAE(\gamma,\lambda=0)} = \delta_t^V}$$

With $\lambda = 0$:

$$\hat{A}_t^{GAE(\gamma,0)} = \delta_t^V = r_t + \gamma V(s_{t+1}) - V(s_t)$$

$$\mathbb{E}_\pi\left[\hat{A}_t^{GAE(\gamma,0)} \mid s_t, a_t\right] = \mathbb{E}_\pi\left[r_t + \gamma V(s_{t+1}) - V(s_t) \mid s_t, a_t\right]$$

$$= \mathbb{E}_\pi\left[r_t + \gamma V(s_{t+1}) \mid s_t, a_t\right] - \mathbb{E}_\pi\left[V(s_t) \mid s_t, a_t\right]$$

$$= Q(s_t, a_t) - V(s_t)$$

True advantage is given by:

$$A^\pi(s_t, a_t) = Q^\pi(s_t, a_t) - V^\pi(s_t)$$

$\hat{A}_t^{GAE(\gamma,0)}$ is an unbiased estimator of the true advantage if and only if:

$$V^\pi(s_t) = V(s_t)$$
$$V^\pi(s_{t+1}) = V(s_{t+1})$$

Then

$$\mathbb{E}_\pi\left[\hat{A}_t^{GAE(\gamma,0)} \mid s_t, a_t\right] = A^\pi(s_t, a_t)$$

i.e. The data-roll out must be collected using an on-policy approach.

© when $\lambda=1$

$$\hat{A}_t^{\text{GAE}(\gamma,\lambda=1)} = \hat{A}_t^{\text{GAE}(\gamma,1)} = \sum_{l\geq 0}^{\infty} \gamma^l \delta_{t+l}^V$$

In infinite horizon, rollouts generated under policy π

$$\mathbb{E}_{\tau\sim\pi}\left[\hat{A}_t^{\text{GAE}(\gamma,1)} \mid s_t, a_t\right] = \mathbb{E}_{\tau\sim\pi}\left[\sum_{l\geq 0}^{\infty} \gamma^l \delta_{t+l}^V \mid s_t, a_t\right]$$

$$= \mathbb{E}_{\tau\sim\pi}\left[\sum_{l\geq 0}^{\infty} \gamma^l \left(\begin{array}{l} r_{t+l} \\ +\gamma V(s_{t+l+1}) \\ -V(s_{t+l}) \end{array}\right) \mid s_t, a_t\right]$$

$$= \mathbb{E}_{\tau\sim\pi}\left[\begin{array}{l} r_t + \gamma V(s_{t+1}) - V(s_t) \\ +\gamma r_{t+1} + \gamma^2 V(s_{t+2}) - \gamma V(s_{t+1}) \\ +\gamma^2 r_{t+2} + \gamma^3 V(s_{t+3}) - \gamma^2 V(s_{t+2}) \\ +\cdots \end{array} \mid s_t, a_t\right]$$

$$= \mathbb{E}_{\tau\sim\pi}\left[\begin{array}{l} r_t + \gamma r_{t+1} \\ +\gamma^2 r_{t+2} + \cdots \\ -V(s_t) \end{array} \mid s_t, a_t\right]$$

$$= \mathbb{E}_{\tau\sim\pi}\left[\left(\sum_{l\geq 0}^{\infty} \gamma^l r_{t+l}\right) - V(s_t) \mid s_t, a_t\right]$$

$$= \mathbb{E}_{\tau\sim\pi}\left[\sum_{l\geq 0}^{\infty} \gamma^l r_{t+l} \mid s_t, a_t\right] - \mathbb{E}_{\tau\sim\pi}\left[V(s_t) \mid s_t, a_t\right]$$

$$\mathop{\mathbb{E}}_{\tau\sim\pi}\left[\hat{A}_t^{GAE(\gamma,1)} \mid s_t, a_t\right] = Q^\pi(s_t,a_t) - V(s_t)$$

$$= \underbrace{Q^\pi(s_t,a_t) - V^\pi(s_t)} + V^\pi(s_t) - V(s_t)$$

$$\downarrow$$

$$A^\pi(s_t,a_t)$$

$$\therefore$$

$$\mathop{\mathbb{E}}_{\tau\sim\pi}\left[\hat{A}_t^{GAE(\gamma,1)} \mid s_t, a_t\right] = A^\pi(s_t,a_t) + V^\pi(s_t) - V(s_t)$$

To show the resulting policy gradient estimate is unbiased:

$$\mathop{\mathbb{E}}\left[\hat{A}_t^{GAE(\gamma,1)} \nabla_\theta \log \pi_\theta(a_t|s_t)\right]$$

$$= \mathop{\mathbb{E}}_{(s_t,a_t)}\left[\mathop{\mathbb{E}}\left[\hat{A}_t^{GAE(\gamma,1)} \nabla_\theta \log \pi_\theta(a_t|s_t) \mid s_t, a_t\right]\right]$$

$$= \mathop{\mathbb{E}}_{(s_t,a_t)}\left[\nabla_\theta \log \pi_\theta(a_t|s_t) \cdot \underbrace{\mathop{\mathbb{E}}\left[\hat{A}_t^{GAE(\gamma,1)} \mid s_t, a_t\right]}\right]$$

$$\downarrow$$

$$A^\pi(s_t,a_t) + V^\pi(s_t) - V(s_t)$$

$$= \mathop{\mathbb{E}}_{(s_t,a_t)}\left[\nabla_\theta (\log \pi_\theta(s_t,a_t) A^\pi(s_t,a_t)\right]$$

$$+ \mathop{\mathbb{E}}_{(s_t,a_t)}\left[\nabla_\theta \log \pi_\theta(s_t,a_t)\left(V^\pi(s_t) - V(s_t)\right)\right]$$

To be minimized:

$$\mathbb{E}_{(s_t, a_t)}\left[\nabla_\theta \log \pi_\theta(s_t, a_t)\left(V^\pi(s_t) - V(s_t)\right)\right] \to 0$$

must go to zero

Taking marginal over actions

$$\mathbb{E}_{s_t}\left[\mathbb{E}_{a_t \sim \pi(\cdot|s_t)}\left[\nabla_\theta \log \pi_\theta(a_t|s_t)\left(V^\pi(s_t) - V(s_t)\right) | s_t\right]\right]$$

$$= \mathbb{E}_{s_t}\left[\left(V^\pi(s_t) - V(s_t)\right) \cdot \mathbb{E}_{a_t \sim \pi(\cdot|s_t)}\left[\nabla_\theta \log \pi_\theta(a_t|s_t) \mid s_t\right]\right]$$

$$\nabla_\theta \log x(\theta) = \frac{1}{x(\theta)} \cdot \nabla_\theta x(\theta)$$

$$= \mathbb{E}_{s_t}\left[\left(V^\pi(s_t) - V(s_t)\right) \cdot \sum_{a_t} \pi_\theta(a_t|s_t) \cdot \frac{1}{\pi_\theta(a_t|s_t)} \cdot \nabla_\theta \pi_\theta(a_t|s_t)\right]$$

$$= \mathbb{E}_{s_t}\left[\left(V^\pi(s_t) - V(s_t)\right) \cdot \sum_{a_t} \nabla_\theta \pi_\theta(a_t|s_t)\right]$$

$$= \mathbb{E}_{s_t}\left[\left(V^\pi(s_t) - V(s_t)\right) \cdot \nabla_\theta \sum_{a_t} \pi_\theta(a_t|s_t)\right]$$

$$= \mathbb{E}_{s_t}\left[\left(V^\pi(s_t) - V(s_t)\right) \cdot \nabla_\theta(1)\right]$$

$$= \mathbb{E}_{s_t}\left[\left(V^\pi(s_t) - V(s_t)\right) \cdot 0\right] = 0$$

$$\therefore \mathbb{E}\left[\hat{A}_t^{GAE(\gamma,1)} \nabla_\theta \log \pi_\theta(a_t|s_t)\right]$$

$$= \mathbb{E}_{(s_t,a_t)}\left[\nabla_\theta \log \pi_\theta(a_t|s_t) A^{\pi}(a_t,s_t)\right]$$

$$+ \mathbb{E}_{(s_t,a_t)}\left[\nabla_\theta \log \pi_\theta(a_t|s_t)\left(V^{\pi}(s_t) - V(s_t)\right)\right]$$

$$= \mathbb{E}_{(s_t,a_t)}\left[\nabla_\theta \log \pi_\theta(a_t|s_t) \cdot A^{\pi}(a_t,s_t)\right]$$

Why does $\lambda=1$ lead to noisy policy-gradients:

With $\lambda=1$, the GAE advantage uses the full roll-out from monte-carlo. The full roll-out has higher variance as variance increases with the trajectory length or roll-out length. Then:

$$\hat{A}_t^{GAE(\gamma,1)} = \left(\sum_{l\geq 0}^{\infty} \gamma^l r_{t+l}\right) - V(s_t)$$

This removes all intermediate bootstrapped value predictions $V(s_{t+l})$ and the estimator depends on the complete sum of future empirical rewards over the entire episode trajectory.

Q2.] The clipped Surrogate Objective As a Pessimistic Lower Bound

$$r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{old}}(a_t|s_t)}$$

$$L^{CLIP}(\theta) = \hat{E}_t\left[\min\left(r_t(\theta)\hat{A}_t,\ clip(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t\right)\right]$$

$$\epsilon \in [0,1)$$

(a) Case 1: $\hat{A}_t > 0$; $r_t(\theta) > 1+\epsilon$

- $L^{CLIP}(\theta) = (1+\epsilon)\hat{A}_t$
- $\nabla_\theta L^{CLIP}(\theta) = \nabla_\theta\left((1+\epsilon)\hat{A}_t\right)$
  $= 0$

Case 2: $\hat{A}_t > 0$; $r_t(\theta) \leq 1+\epsilon$

- $L^{CLIP}(\theta) = r_t(\theta)\hat{A}_t$
- $\nabla_\theta L^{CLIP}(\theta) = \nabla_\theta\left(r_t(\theta)\hat{A}_t\right) = \hat{A}_t \nabla_\theta r_t(\theta)$
  $= \hat{A}_t \cdot \frac{\nabla_\theta \pi_\theta(a_t|s_t)}{\pi_{\theta_{old}}(a_t|s_t)}$

Case 3: $\hat{A}_t < 0$; $r_t(\theta) < 1-\epsilon$

- $L^{CLIP}(\theta) = (1-\epsilon)\hat{A}_t$
- $\nabla_\theta L^{CLIP}(\theta) = \nabla_\theta\left((1-\epsilon)\hat{A}_t\right)$
  $= 0$

Case 4: $\hat{A}_t < 0 \gg 1-\epsilon$

- $L^{CLIP}(\theta) = r_t(\theta)\hat{A}_t$
- $\nabla_\theta L^{CLIP}(\theta) = \nabla_\theta \left(r_t(\theta)\hat{A}_t\right)$
  $= \hat{A}_t \nabla_\theta r_t(\theta)$
  $= \hat{A}_t \dfrac{\nabla_\theta \pi_\theta(a_t|s_t)}{\pi_{\theta_{old}}(a_t|s_t)}$

(6)

→ $\hat{A}_t > 0$

$clip(r_t(\theta), 1-\epsilon, 1+\epsilon) \leq 1+\epsilon$

$\therefore clip(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t \leq (1+\epsilon)\hat{A}_t$

$\therefore \min(r_t(\theta)\hat{A}_t, clip(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t) \leq r_t(\theta)\hat{A}_t$

→ $\hat{A}_t < 0$

$clip(r_t(\theta), 1-\epsilon, 1+\epsilon) \geq 1-\epsilon$

$\therefore clip(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t \leq (1-\epsilon)\hat{A}_t$

- If $r_t(\theta) \gg 1-\epsilon$

$clip(r_t(\theta), 1-\epsilon, 1+\epsilon) = r_t(\theta)$

$\therefore \min(r_t(\theta)\hat{A}_t, r_t(\theta)\hat{A}_t) = r_t(\theta)\hat{A}_t \leq r_t(\theta)\hat{A}_t$

- If $r_t(\theta) < 1-\epsilon$

$clip(r_t(\theta), 1-\epsilon, 1+\epsilon) = 1-\epsilon$

$\therefore \min(r_t(\theta)\hat{A}_t, (1-\epsilon)\hat{A}_t) = (1-\epsilon)\hat{A}_t \leq r_t(\theta)\hat{A}_t$

$$\therefore \quad L^{CLIP}(\theta) \leq r_t(\theta)\hat{A}_t \;;\; \forall\, r_t(\theta) > 0 \text{ and } \hat{A}_t \in \mathbb{R}$$

The $L^{CLIP}(\theta)$ acts as a pessimistic lower bound due to the pointwise minimum between the unclipped objective and the clipped objective.

For good actions ($\hat{A}_t > 0$), the $L^{CLIP}(\theta)$ caps the reward at $(1+\epsilon)\hat{A}_t$ and its min operator chooses this lower bound preventing inflation of the unclipped objective

For bad actions ($\hat{A}_t < 0$), the $L^{CLIP}(\theta)$ and its min operator select the smallest of $r_t(\theta)\hat{A}_t$ and $(1-\epsilon)\hat{A}_t$ allowing for policy update by selecting the more negative value of the two.

2. CONTINUOUS CONTROL IMPLEMENTATION IN MUJOCO

[06]

b) GAE Trade-off Ablation

Figure 2 shows a clear trade-off in convergence based on the value of λ. The default value of $\lambda = 0.95$ converges fastest to the maximum return of 1000 and is stable at that point.

The 1-step TD baseline ($\lambda = 0.0$) converges slowly, reaching the maximum return much later than for $\lambda = 0.95$. The evaluation for this 1-step TD baseline is relatively stable.

The full monte-carlo rollout ($\lambda=1.0$) is the slowest converging much more later than the 1-step TD baseline ($\lambda=0.0$) and the default value ($\lambda=0.95$). This monte-carlo estimation also shows high variance in its evaluation before it finally converges.

These results confirm that $\lambda=0.95$ provides the best balance It reduces variance relative to Monte Carlo while retaining enough information across steps to avoid bias that slows down convergence.

c) Policy Clipping Ablation

Disabling Clipping as shown in figure 3 produces instability throughout the training process. In contrast, the clipped policy converges smoothly to 1000 and remains stable.

In figure 4, the KL-divergence plots makes the failure mode clear. The unclipped policy has large KL values compared to that of the clipped policy which stays at zero. This shows that without clipping, the optimizer takes excessively large policy update steps, the new policy drifts far from the old one in a single gradient step destroying previously learned behavior.

In continuous action spaces, the Gaussian policy's mean and variance can change significantly in a single update, causing the agent to suddenly explore regions of action space it had learned to avoid. The clipped surrogate objective prevents this by zeroing out gradient whenever the probability ratio is outside $[1-\varepsilon, 1+\varepsilon]$, effectively enforcing a trust region without requiring a constraint solver.